## Q1. Loss Function

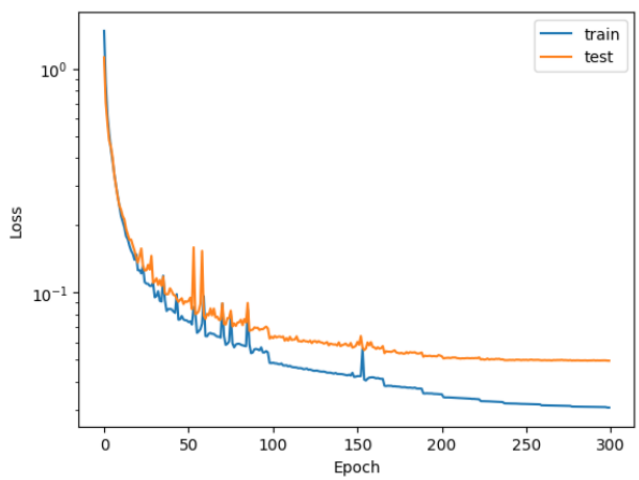
$$\mathcal{L} = E_{x_0,\,t,\,\epsilon}\ \|\epsilon - \epsilon_\theta(x_t, t, y)\|^2 \tag{1}$$

where:

- $x_0$ is the clean SDF,
- $x_t$ is the noisy SDF at timestep $t$,
- $\epsilon$ is the Gaussian noise added to $x_0$,
- $\epsilon_\theta(\cdot)$ is the model prediction,
- $y$ is the conditioning stress–strain curve.

Thus, the model learns to predict and remove noise.

## Q2. Loss History

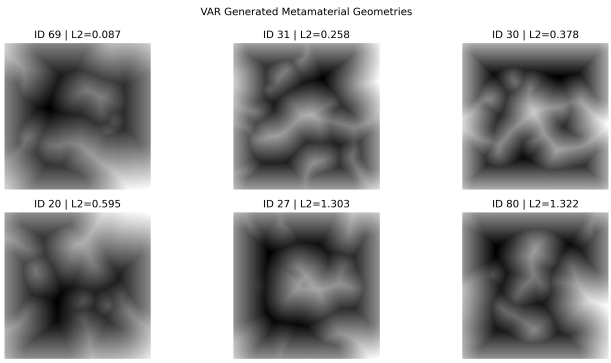


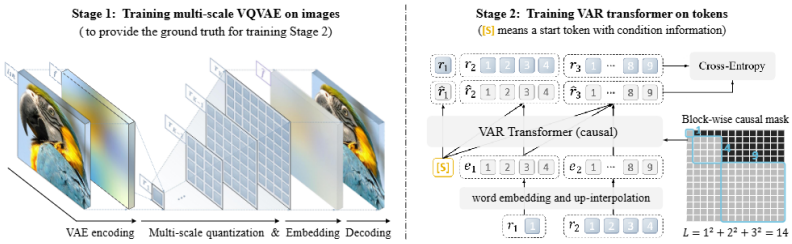

The small gap between training and validation loss suggests limited overfitting and good generalization.

## Q3. Biggest Challenge

The main challenge in the VAR-based approach is that it requires training a VQ-VAE to discretize the continuous SDF space into tokens, which currently

results in poor-quality reconstructions and consequently degrades the generated samples (see Fig. ).




The original VAR framework relies on hierarchical resolution prediction, where images are generated progressively from low to high resolution through a multi-scale token generation process, as illustrated in Fig. .